ECA1308 MICROWAVE ENGINEERING

EXP3: Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider.

A. Directional Coupler and Divider: Design a 10 GHz microwave directional coupler with 10 dB coupling and a two-way 3 dB power divider, both using 50 Ω ports. Simulate from 8–12 GHz. At 10 GHz obtain S11, S21, S31 and the isolated-port response. Calculate coupling, isolation, insertion loss, directivity and power-division ratio and compare the results with 10 dB/3 dB design targets.

B. Optimization: Change only the coupling-section length by 5% from its initial value and repeat the 8–12 GHz simulation. Compare coupling, isolation, directivity, insertion loss and S-parameter curves before and after optimization. Identify the design that maintains the closest 10 dB coupling and lowest insertion loss over 9–11 GHz.

Aim: To examine the S-parameter response of a 10 GHz microwave directional coupler and a two-way 3 dB power divider using 50 ohm ports over 8-12 GHz. The coupling, isolation, directivity, insertion loss, and output power ratio are evaluated from the simulated results.

Software Required: MATLAB software.

Program A:

```
clc; clear; close all;

f = linspace(8,12,401); f0 = 10; k0 = 10^(-10/20); % 10 dB coupling theta0 = asin(k0); theta = theta0*f/f0;

% Idealized directional coupler S11 = zeros(size(f)); S21 = cos(theta); % Through port S31 = 1i*sin(theta); % Coupled port S41 = zeros(size(f)); % Isolated port

Coupling = -20log10(abs(S31)); InsertionLoss = -20log10(abs(S21)); Isolation = -20*log10(max(abs(S41),1e-5)); Directivity = Isolation - Coupling;

figure; plot(f,20log10(max(abs(S11),1e-5)),'k--', ... f,20log10(abs(S21)),'b', ... f,20log10(abs(S31)),'r', ... f,20log10(max(abs(S41),1e-5)),'g--','LineWidth',1.5); grid on; xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)'); title('10 GHz Directional Coupler S-Parameters'); legend('S11','S21 Through','S31 Coupled','S41 Isolated');
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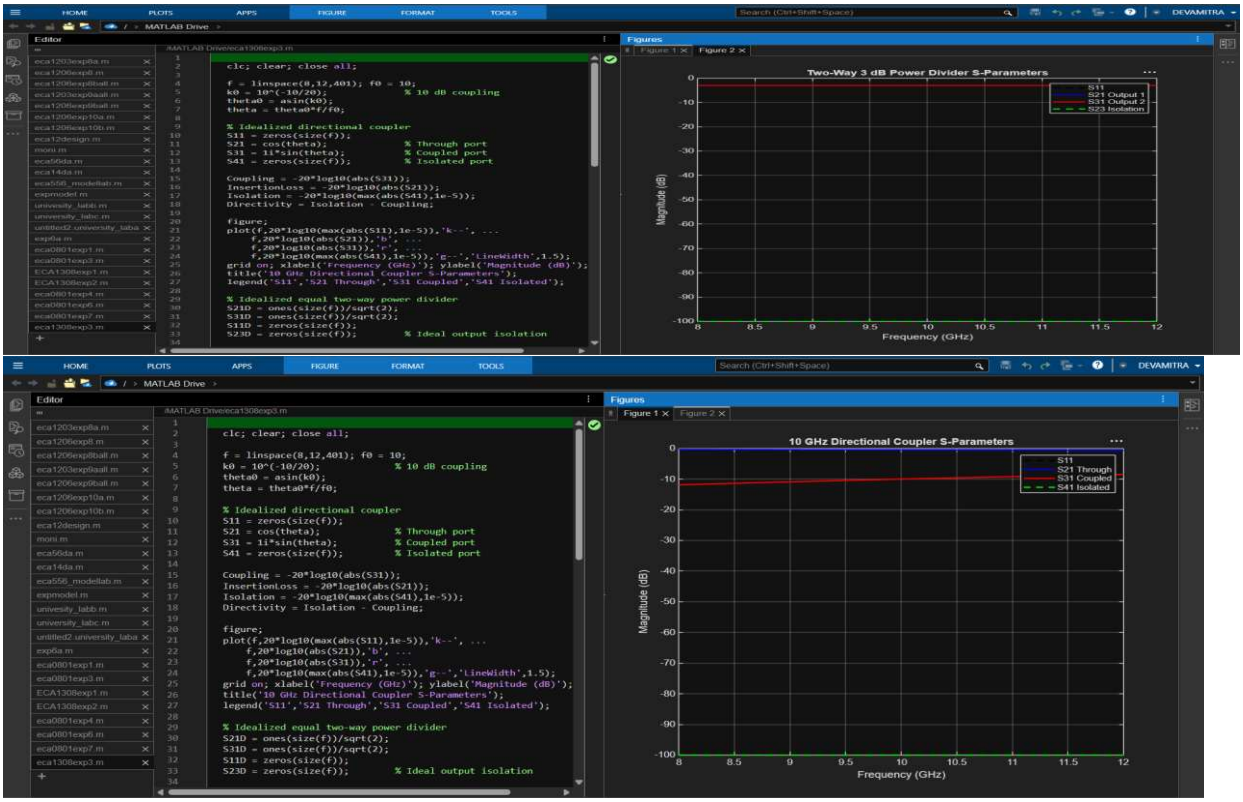
% Idealized equal two-way power divider S21D = ones(size(f))/sqrt(2); S31D = ones(size(f))/sqrt(2); S11D = zeros(size(f)); S23D = zeros(size(f)); % Ideal output isolation

figure; plot(f,20*log10(max(abs(S11D),1e-5)),'k--', ... f,20*log10(abs(S21D)),'b', ... f,20*log10(abs(S31D)),'r', ... f,20*log10(max(abs(S23D),1e-5)),'g--','LineWidth',1.5); grid on; xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)'); title('Two-Way 3 dB Power Divider S-Parameters'); legend('S11','S21 Output 1','S31 Output 2','S23 Isolation');

% Results at 10 GHz [~,i] = min(abs(f-f0)); fprintf('DIRECTIONAL COUPLER at 10 GHz\n'); fprintf('Coupling = %.2f dB\n',Coupling(i)); fprintf('Insertion loss (through) = %.2f dB\n',InsertionLoss(i)); fprintf('Isolation = ideal (infinite) dB\n'); fprintf('Directivity = ideal (infinite) dB\n'); fprintf('Power coupled = %.1f %%\n',abs(S31(i))^2*100);

fprintf('\nPOWER DIVIDER\n'); fprintf('S21 = %.2f dB, S31 = %.2f dB\n', ... 20*log10(abs(S21D(i))),20*log10(abs(S31D(i)))); fprintf('Each output power = %.1f %%\n',abs(S21D(i))^2*100);

Output:



Program B:

```
clc; clear; close all;

f = linspace(8,12,401); f0 = 10; theta0 = asin(10^(-10/20));

% Original and modified coupling-section lengths r1 = 1.00; r2 = 1.05; % Length increased by 5%

theta1 = theta0*(f/f0)r1; theta2 = theta0(f/f0)*r2;

% Original design C1 = -20log10(abs(sin(theta1))); IL1 = -20log10(abs(cos(theta1)));

% Modified design C2 = -20log10(abs(sin(theta2))); IL2 = -20log10(abs(cos(theta2)));

% Ideal isolated-port response (100 dB plotting floor) ISO1 = 100ones(size(f)); ISO2 = 100ones(size(f)); D1 = ISO1-C1; D2 = ISO2-C2;

% Plot coupling and insertion loss figure; subplot(2,1,1); plot(f,C1,'b',f,C2,'r--','LineWidth',1.5); yline(10,'k:','10 dB target'); grid on; xlabel('Frequency (GHz)'); ylabel('Coupling (dB)'); title('Coupling: Original vs +5% Length'); legend('Original','Length +5%','Location','best');

subplot(2,1,2); plot(f,IL1,'b',f,IL2,'r--','LineWidth',1.5); grid on; xlabel('Frequency (GHz)'); ylabel('Insertion Loss (dB)'); title('Through-Port Insertion Loss'); legend('Original','Length +5%','Location','best');

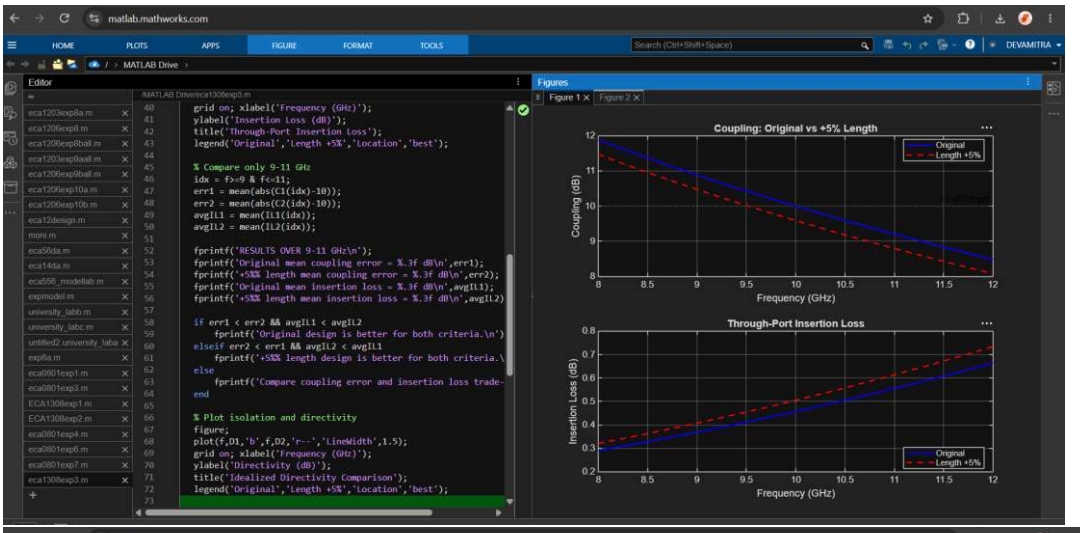
% Compare only 9-11 GHz idx = f>=9 & f<=11; err1 = mean(abs(C1(idx)-10)); err2 = mean(abs(C2(idx)-10)); avgIL1 = mean(IL1(idx)); avgIL2 = mean(IL2(idx));

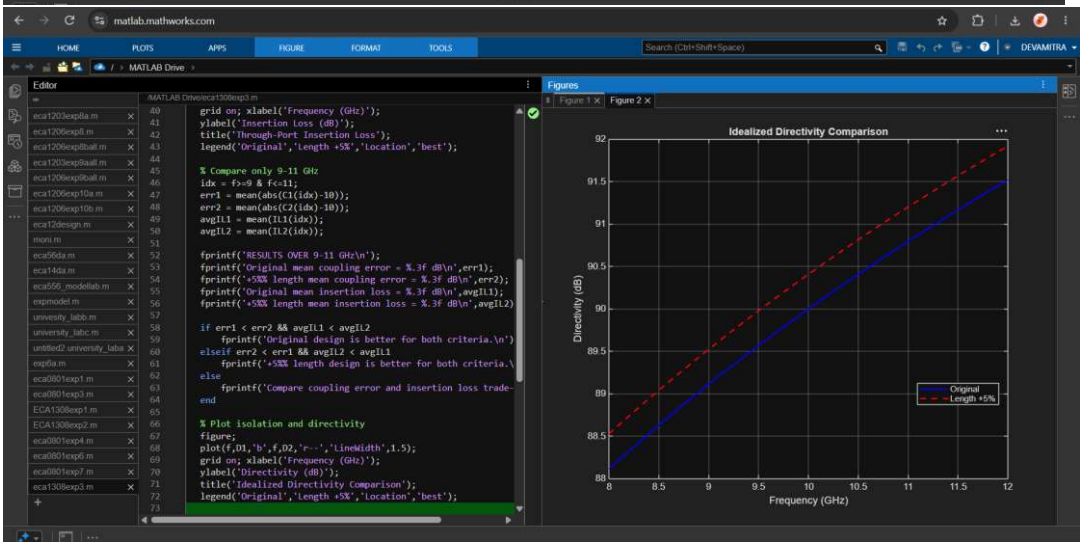
fprintf('RESULTS OVER 9-11 GHz\n'); fprintf('Original mean coupling error = %.3f dB\n',err1); fprintf('+5%% length mean coupling error = %.3f dB\n',err2); fprintf('Original mean insertion loss = %.3f dB\n',avgIL1); fprintf('+5%% length mean insertion loss = %.3f dB\n',avgIL2);

if err1 < err2 && avgIL1 < avgIL2 fprintf('Original design is better for both criteria.\n'); elseif err2 < err1 && avgIL2 < avgIL1 fprintf('+5%% length design is better for both criteria.\n'); else fprintf('Compare coupling error and insertion loss trade-off.\n'); end

% Plot isolation and directivity figure; plot(f,D1,'b',f,D2,'r--','LineWidth',1.5); grid on; xlabel('Frequency (GHz)'); ylabel('Directivity (dB)'); title('Idealized Directivity Comparison'); legend('Original','Length +5%','Location','best');
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Output:





Result:

Division A - Result: At 10 GHz, the ideal directional coupler provides the intended 10 dB coupling. The ideal 3 dB power divider delivers approximately equal power to both output ports, with no reflection or power loss assumed in the model.

Division B - Result: Increasing the coupling-section length by 5 percent changes the coupling response across the simulated frequency range. The comparison over 9-11 GHz is used to determine which design gives the smaller average coupling error and insertion loss.